

# Isomerism

# Configurational Isomers

Configurational isomers are stereoisomers that cannot be converted into one another by rotation around a single bond.  
(geometric isomers and optical isomers).



**geometric isomers**  
cis-trans isomers

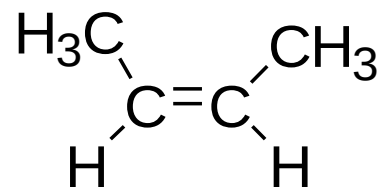
**optical isomers**  
isomers that contain  
chirality centers

# geometric isomers

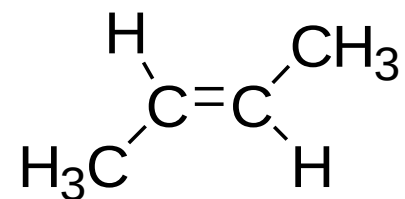
## cis-trans isomers

2-butene :  $\text{CH}_3\text{CH}=\text{CHCH}_3$

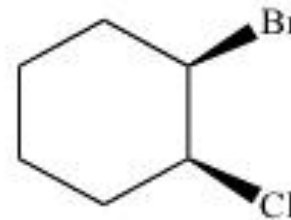
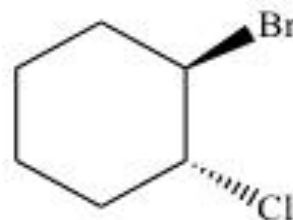
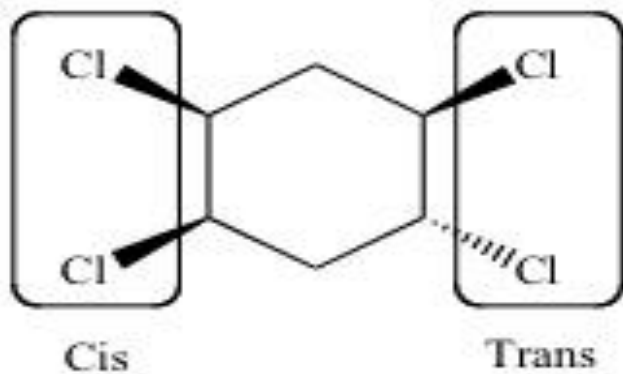
2 geometric isomers



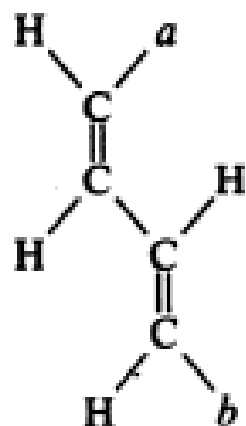
cis-2-butene



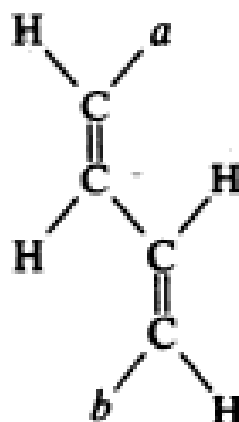
trans-2-butene



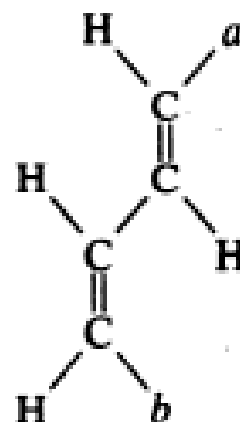
If a compound has two double bonds, four geometrical isomers are possible



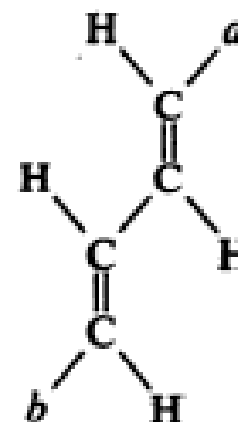
*cis-trans*



*cis-cis*



*trans-cis*



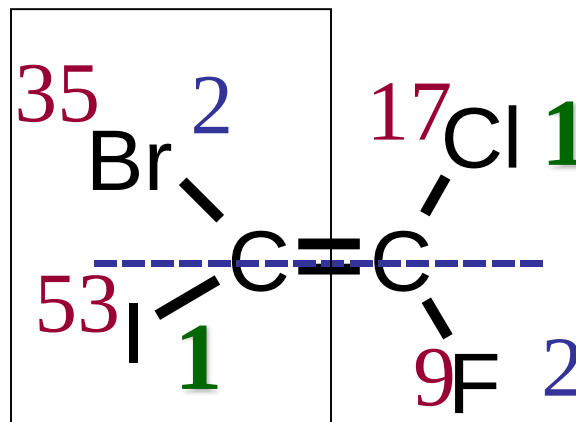
*trans-trans*

# E and Z nomenclature

## E and Z Based on Priority:

### Cahn-Ingold-Prelog Rules

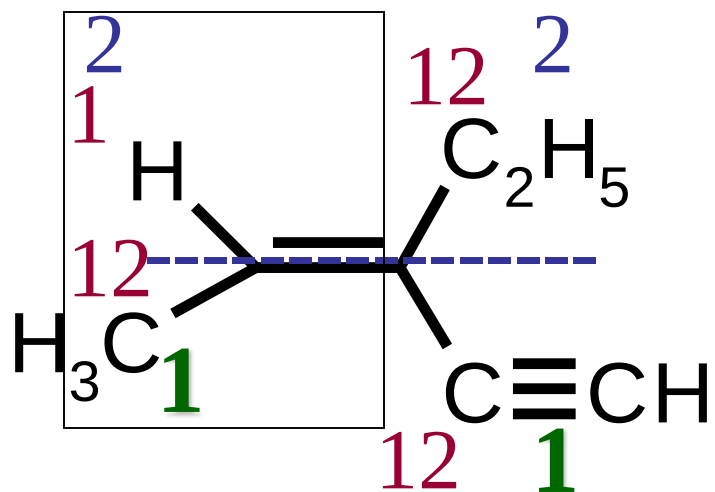
- Assign a priority number to each group attached to the chiral carbon.
- Atom with highest atomic number assigned the highest priority #1.
- In case of ties, look at the next atoms along the chain.
- Double and triple bonds are treated like bonds to duplicate atoms.



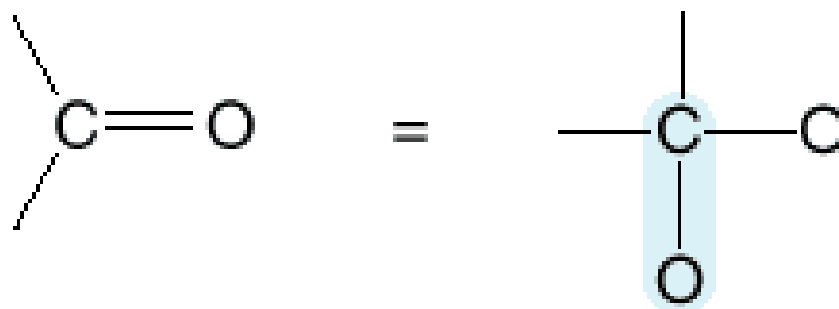
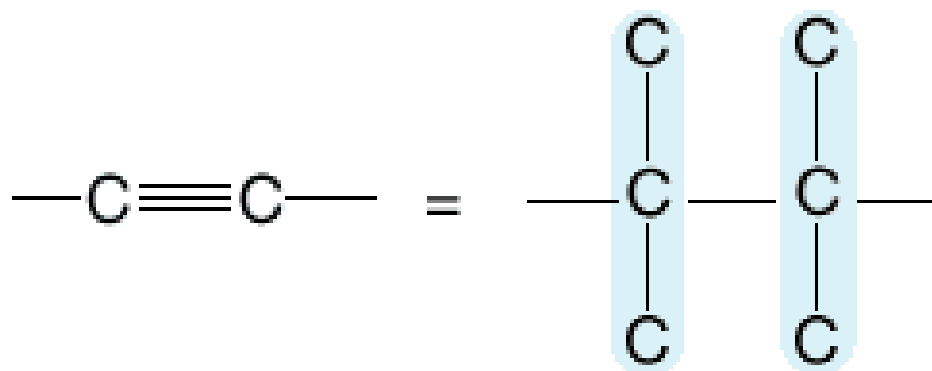
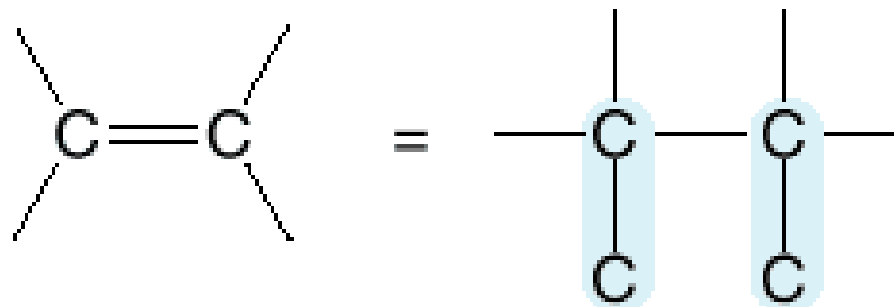
Higher priority at the **opposite** side of pi bond (**E**)

Higher priority at the **same** side of pi bond (**Z**)

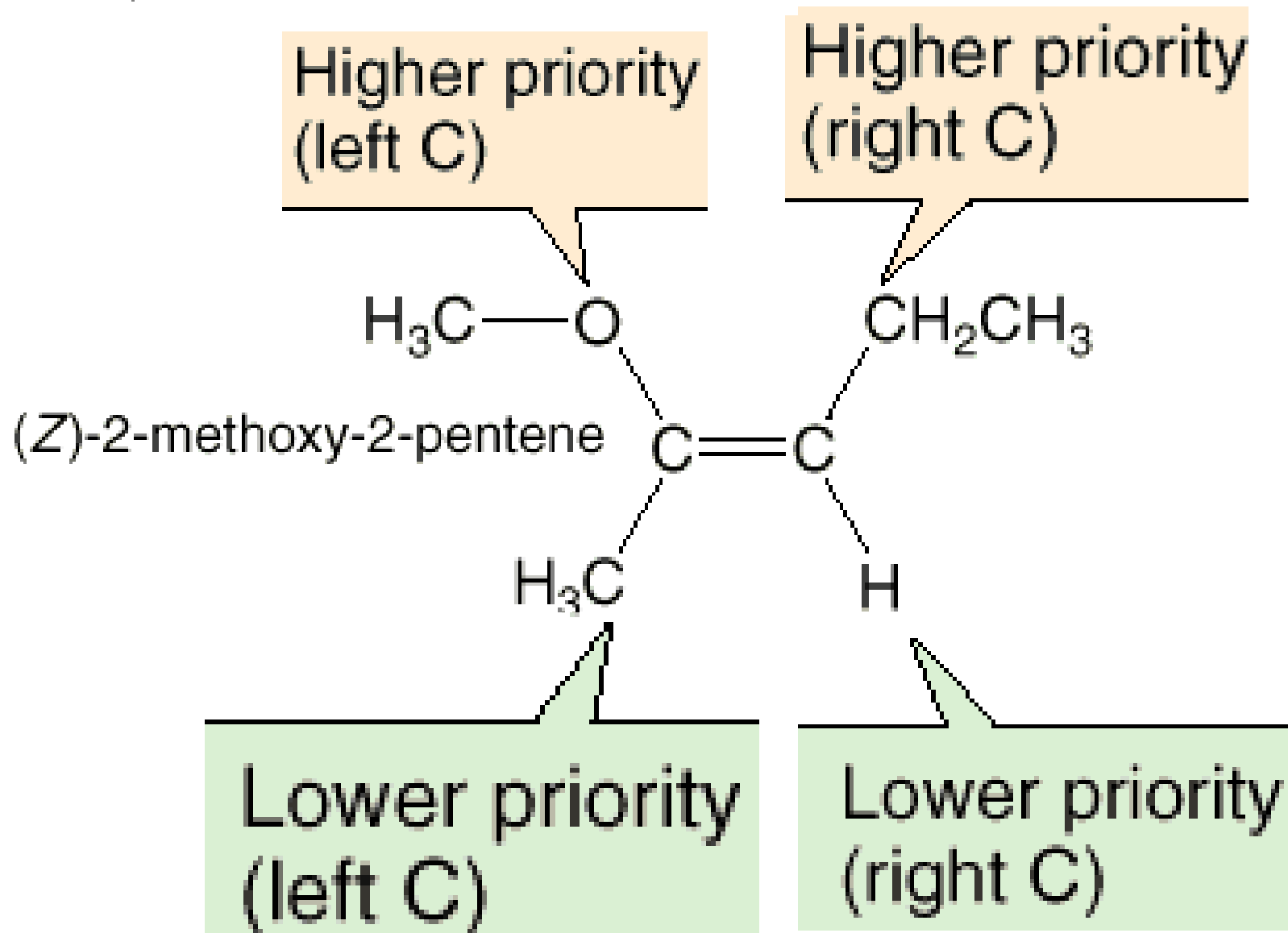
(E)-1-bromo-2-chloro - 2-fluoro-1-iodoethene



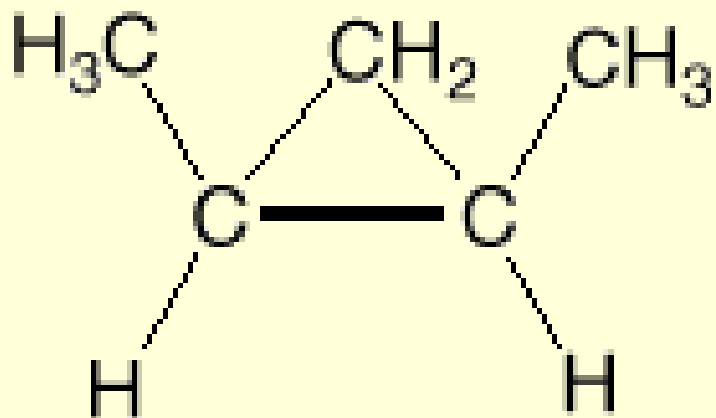
(Z)-3-ethyl-2-pentene-4-yne



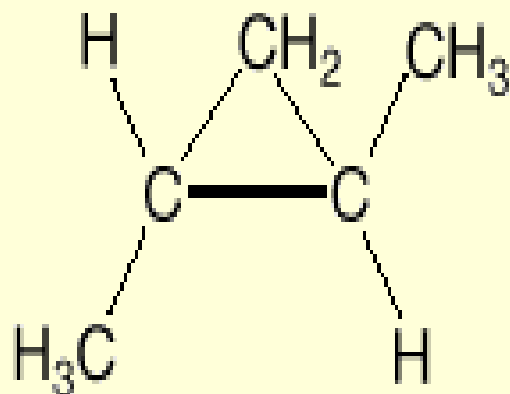
The two high priority groups are on the same side — this molecule is (*Z*)-



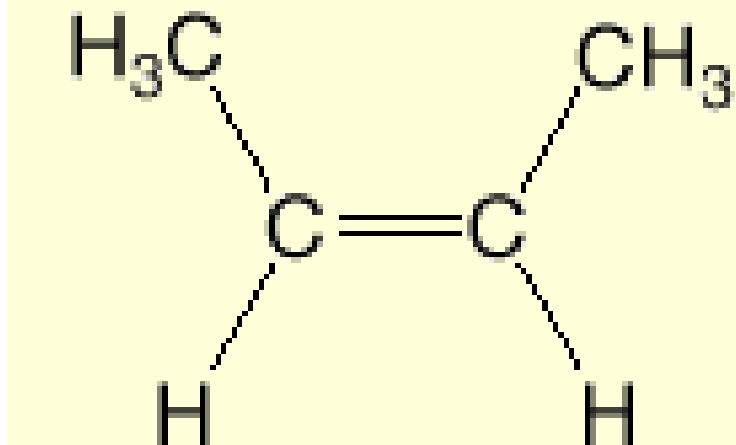
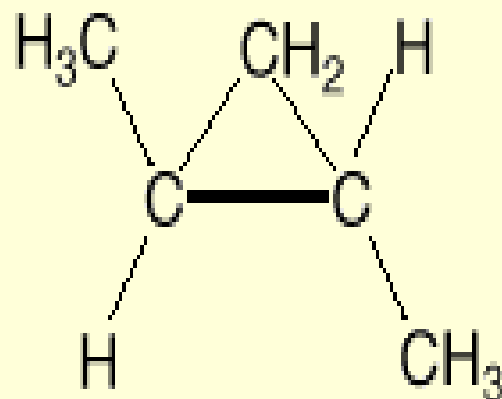
# Geometric Isomers



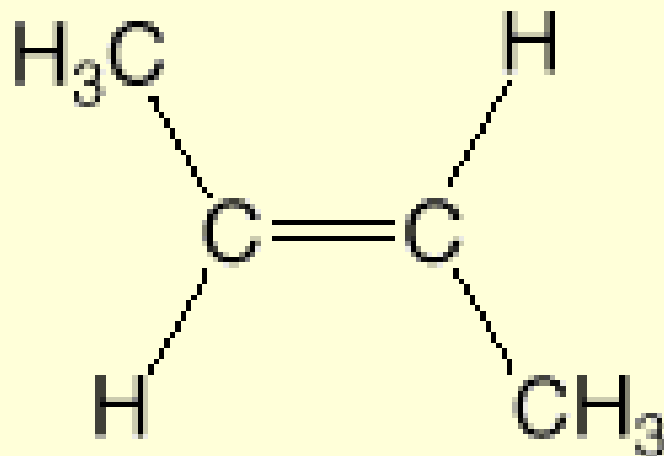
*cis*-1,2-Dimethylcyclopropane



Two representations of  
*trans*-1,2-dimethylcyclopropane



*cis*-2-Butene [or (*Z*)-2-butene]



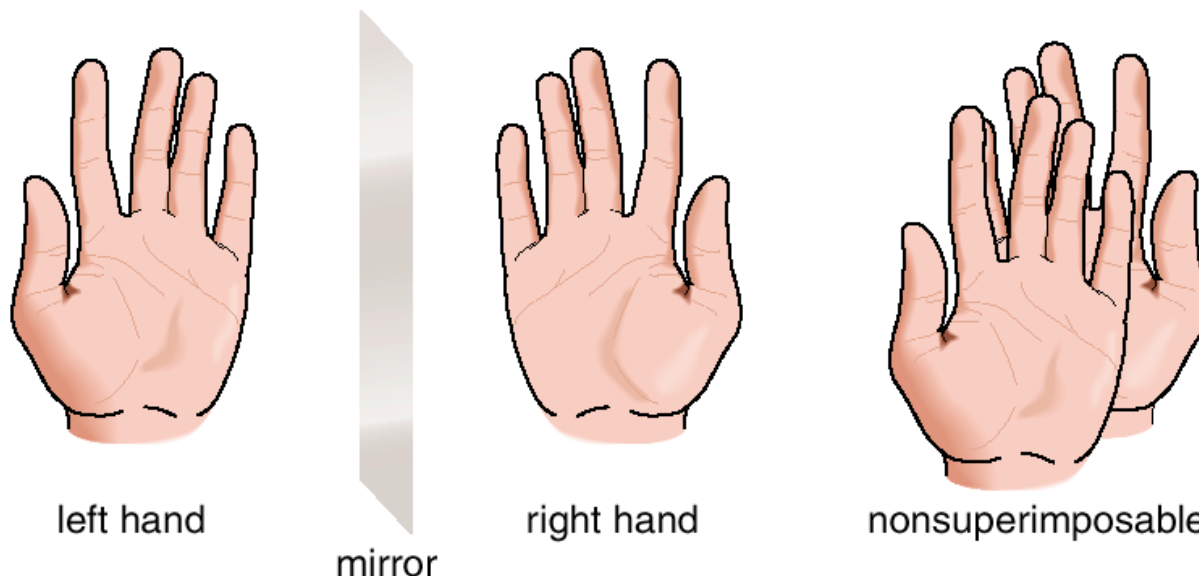
*trans*-2-Butene [or (*E*)-2-butene]



# Optical Isomerism

Some objects are not the same as their mirror images (technically, they have no plane of symmetry)

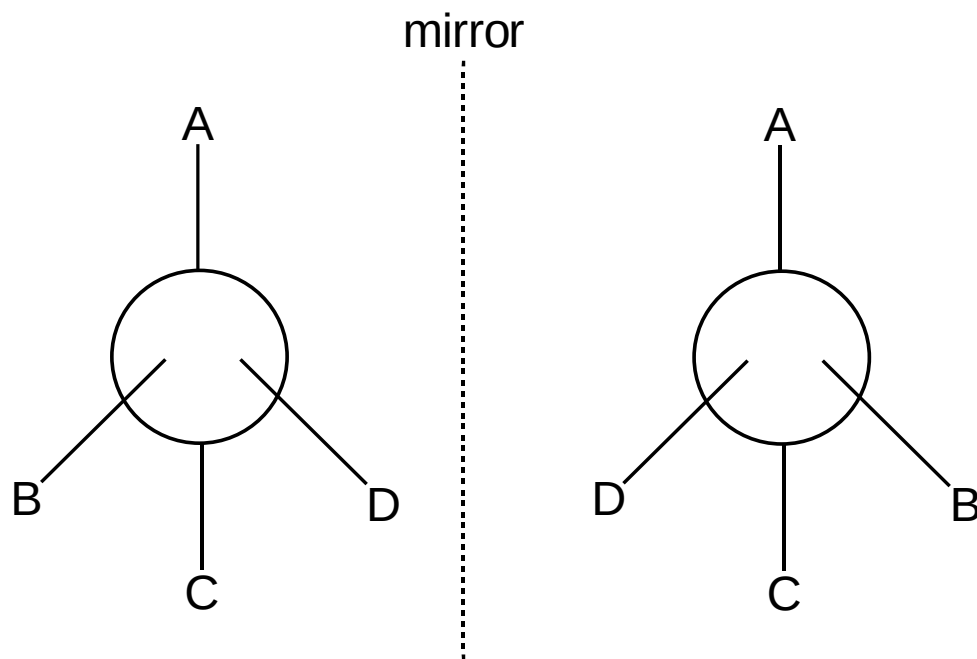
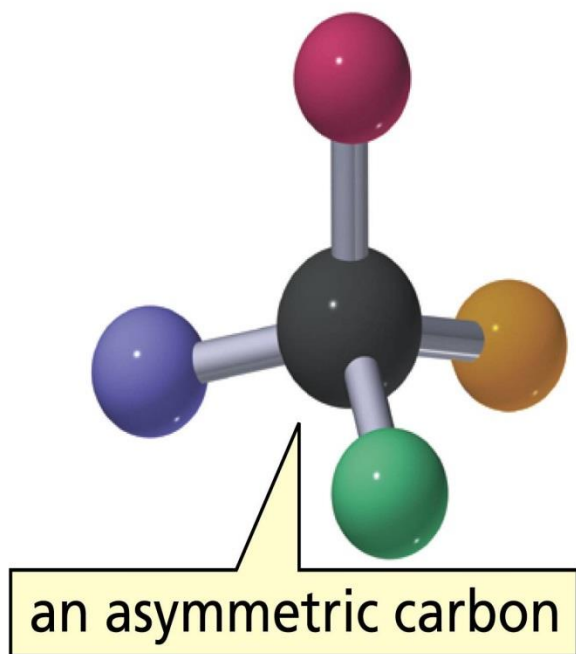
A right-hand glove is different than a left-hand glove. The property is commonly called “handedness”



- A molecule (or object) that is *not* superimposable on its mirror image is said to be *chiral*.

# Chirality

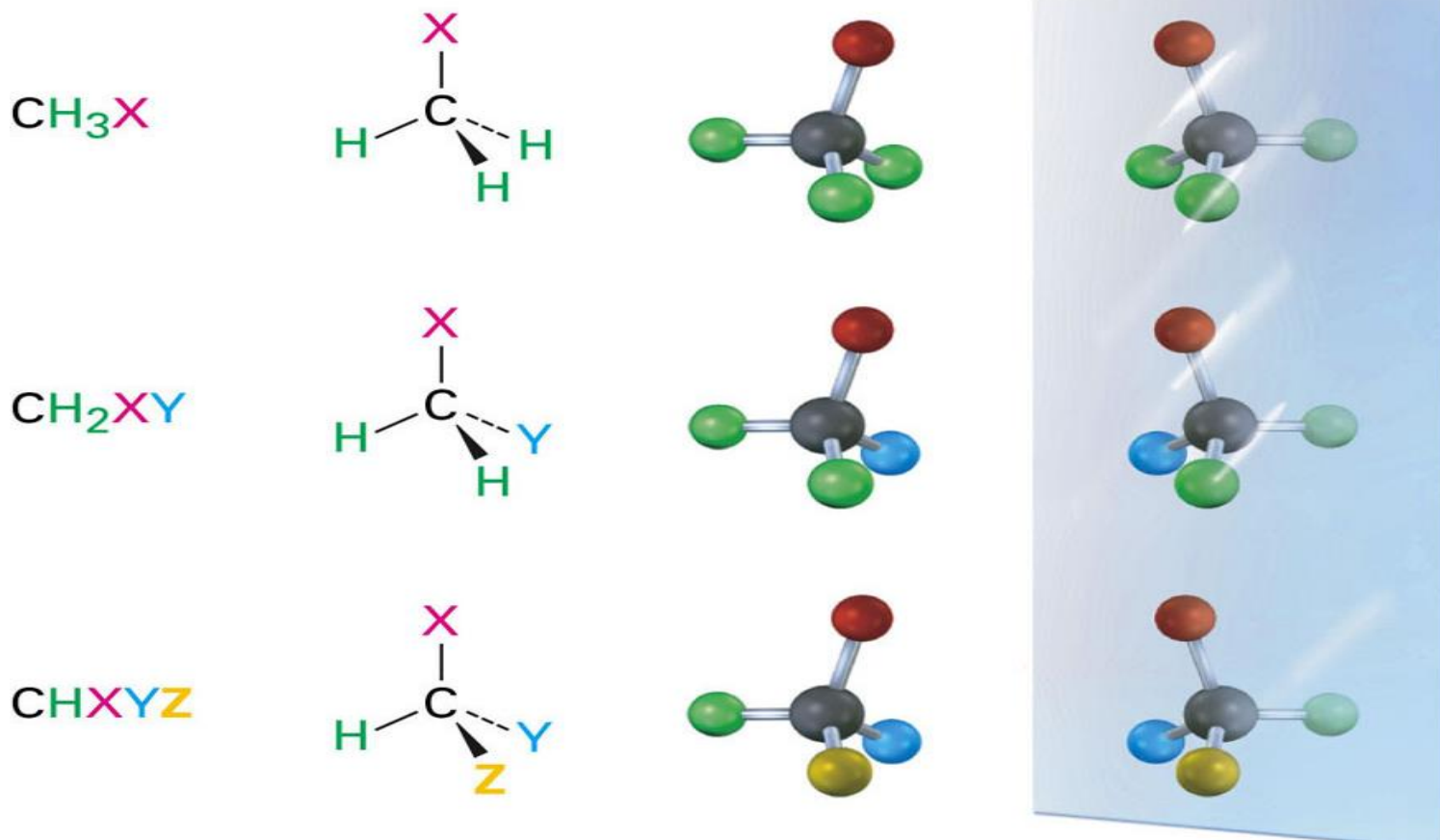
A chiral molecule is a type of molecule that has a non-superposable mirror image. The property is most often the cause of the presence of an asymmetric carbon atom

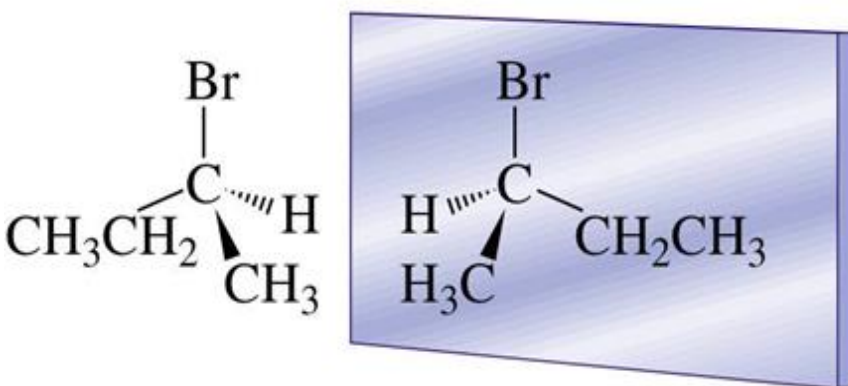


The two are non-superimposable, mirror images. Such isomers are called enantiomers.

# Examples of Enantiomers

- Molecules that have one carbon with 4 different substituents have a non-superimposable mirror image
- **Enantiomers** = non-superimposable mirror image stereoisomers
- Enantiomers have identical physical properties (except for one)
- Build molecular models to see this

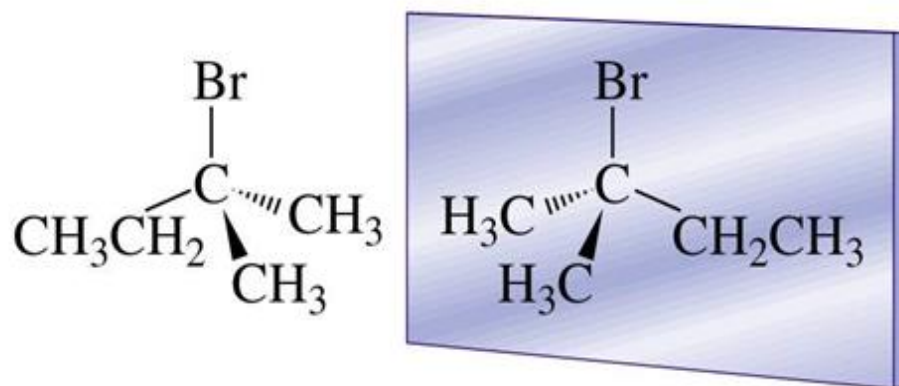




a chiral  
molecule

nonsuperimposable  
mirror image

**enantiomers**



an achiral  
molecule

superimposable  
mirror image

**identical molecules**

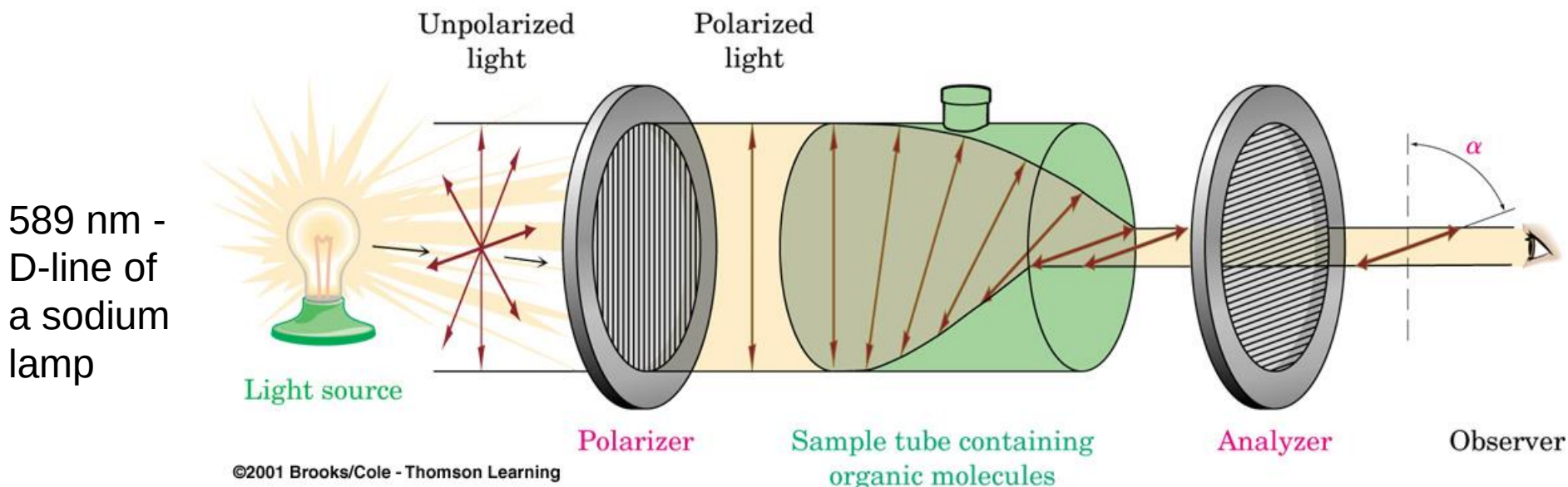
## Optical Activity –

molecules enriched in an enantiomer will rotate plane polarized light are said to be *optically active*. The optical rotation is dependent upon the substance, the concentration, the path length through the sample, and the wavelength of light.

Rotation, in degrees, is  $[\alpha]$

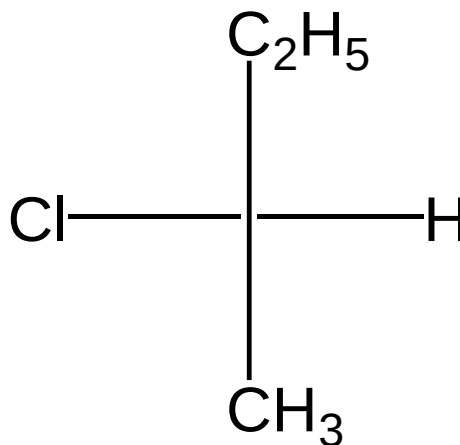
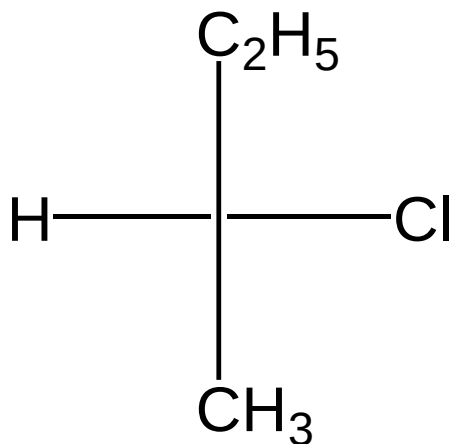
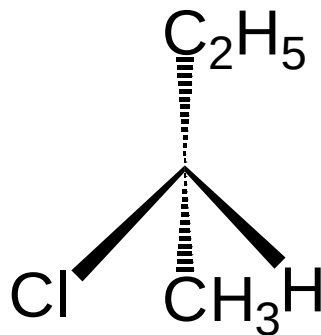
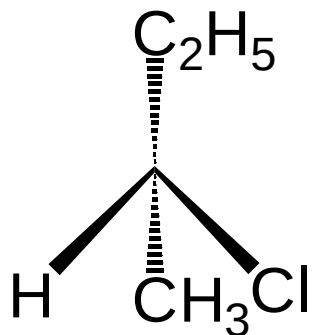
Clockwise (+) = dextrorotatory; Anti-clockwise (-) = levorotatory

## Polarimeter



Plane polarized light: light that oscillates in only one plane

How do we “draw” a chirality centre?

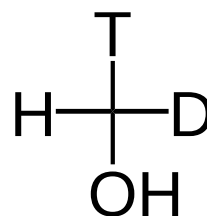
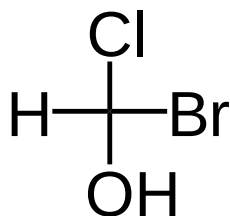


**Fischer Projection**

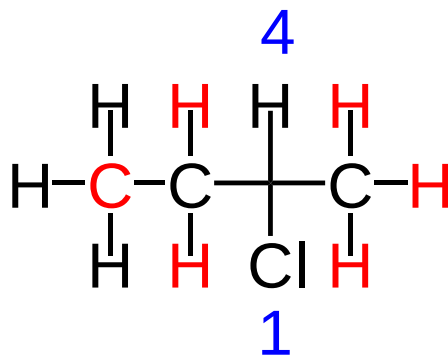
# Cahn - Prelog - Ingold rules for nomenclature

**Step 1:** assign a priority to the 4 atoms or groups of atoms bonded to the tetrahedral stereogenic centre:

1. If the 4 atoms are all different, priority is determined by atomic number. The atom of higher atomic number has the higher priority.

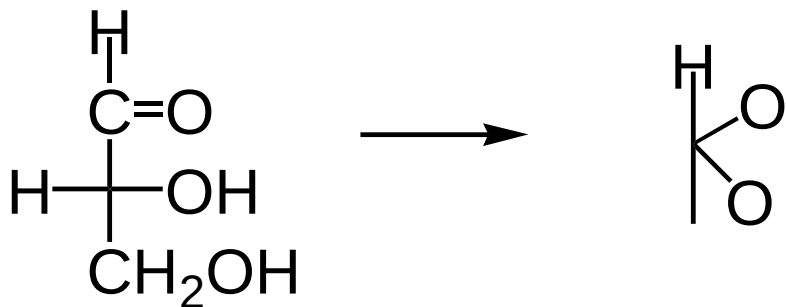


2. If priority cannot be determined by (1), it is determined by a similar comparison of atoms working out from the stereocentre.



In the methyl group, the second atoms are H, H, H whereas in the ethyl group, they are C, H, H. The priority sequence is therefore Cl, C<sub>2</sub>H<sub>5</sub>, CH<sub>3</sub>, H.

3. A double or triple bond to an atom, **A**, is considered as equivalent to two or three single bonds to **A**:

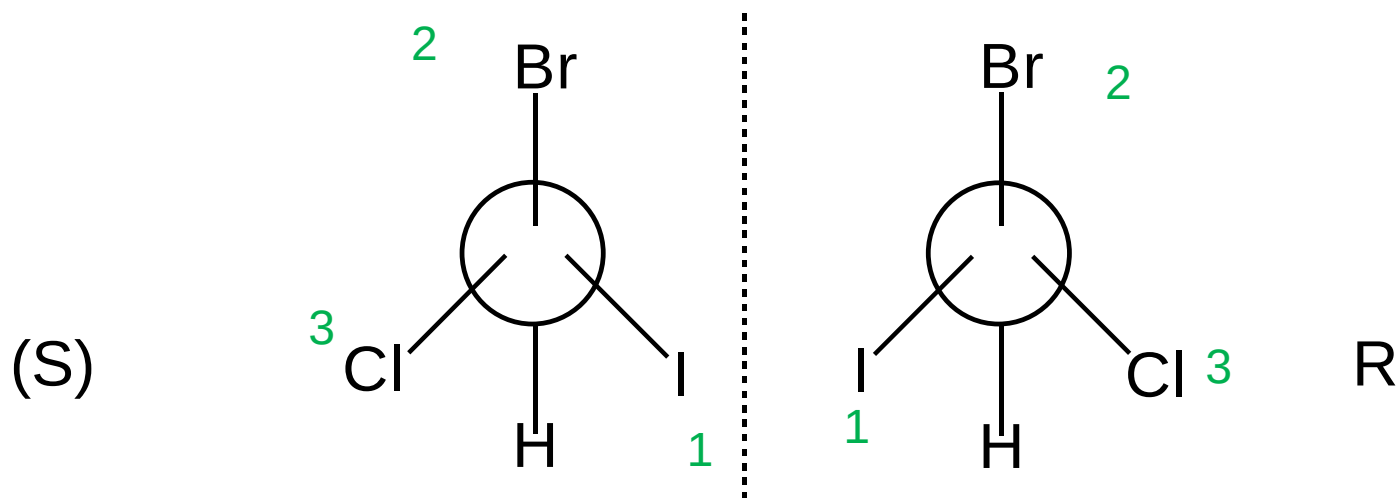


The sequence is therefore -OH, -CHO, -CH<sub>2</sub>OH, -H.



## Step 2

Arrange the molecule so that the group of lowest priority is pointing away from you and observe the arrangement of the remaining groups:

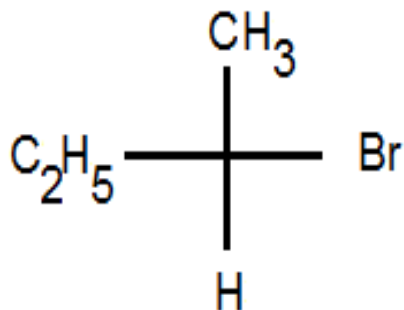


If, on going from the group of highest priority to that of second priority and then to the group of third priority, we go in a clockwise direction, the enantiomer is designated (R).

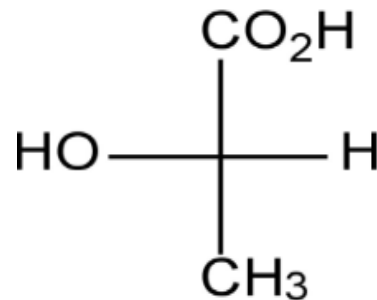
If the direction is counterclockwise, the enantiomer is designated (S).

Thus the complete name for one of the enantiomers of 2-chlorobutane is (R)-2-chlorobutane

# R or S?

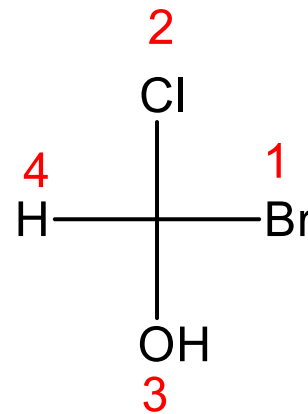
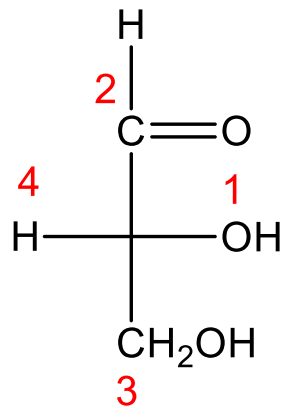
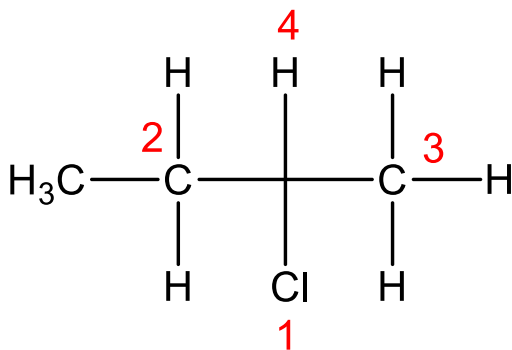


(R)-2-bromobutane



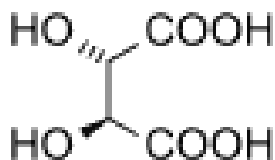
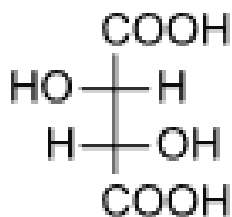
(S)- 2-hydroxy propanoic acid

Try followings

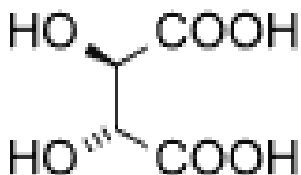
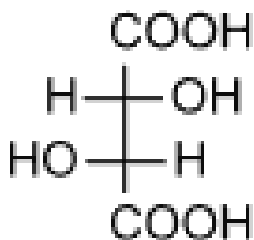


# Diastereomers

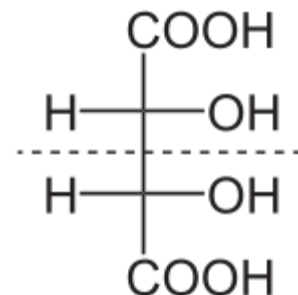
Some optical isomers may rotate the plane of polarization by different amounts, called diastereomers.



L-(+)-tartaric acid  
dextrotartaric acid



D-(-)-tartaric acid  
levotartaric acid

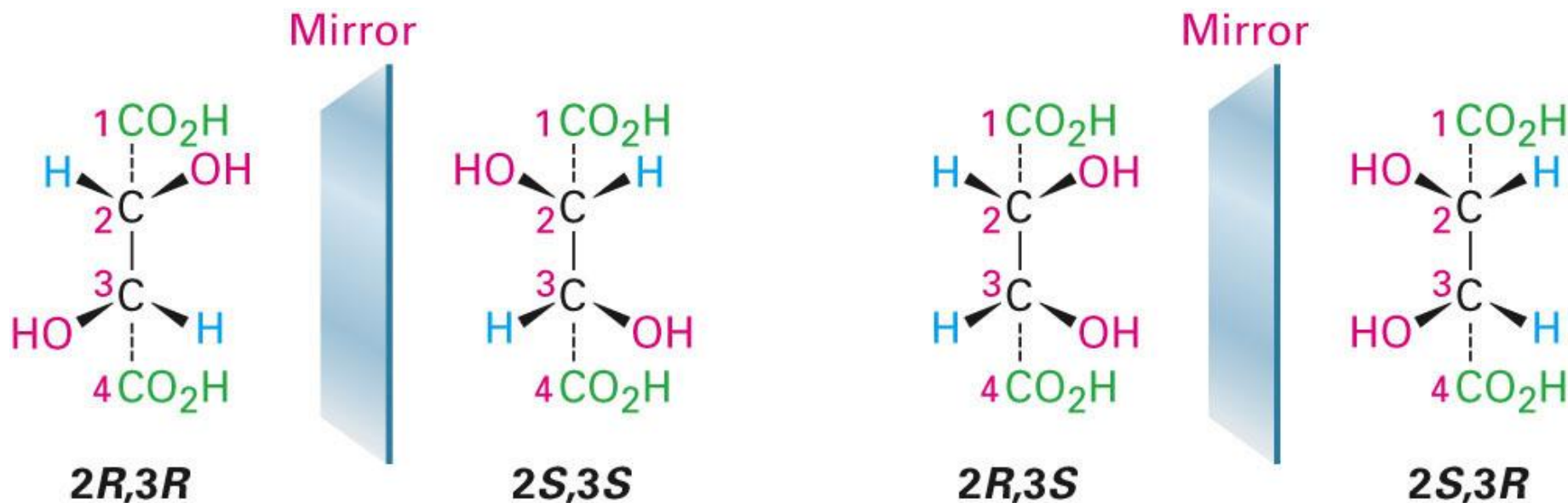


mesotartaric acid

(1:1) DL-tartaric acid "racemic acid"

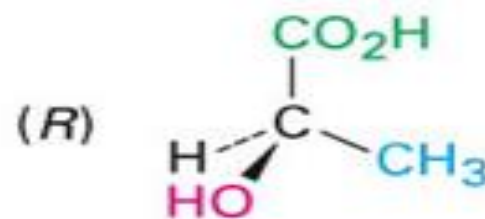
# Meso Compounds

- Tartaric acid has two chiral centers and two diastereomeric forms
- One form is chiral and one is achiral, but both have two chiral centers
- An achiral compound with chiral centers is called a *meso* compound – it has a plane of symmetry
- The two structures on the right in the figure are identical so the compound (**2R, 3S**) is achiral
- Identical substitution on both chiral centers

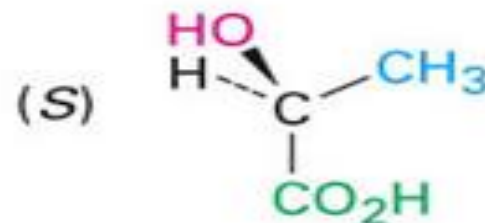


# Racemic Mixtures

- A 50:50 mixture of two chiral compounds that are mirror images does not rotate light – called a **racemic mixture** (named for “racemic acid” that was the double salt of (+) and (-) tartaric acid)



+



**Racemic lactic acid**  
**(50% *R*, 50% *S*)**